

VALIDATION & ACCURACY REPORT

HORIZON 87

ZENITH DE441 State-Vector Validation Against NASA/JPL Horizons
Version 1.0 - June 2026 - Desktop Web Application

REPORT PURPOSE

This report summarizes a matched ZENITH validation run comparing local Horizon 87 CSPICE state vectors against live NASA/JPL Horizons VECTORS references. It is designed as the public accuracy companion to the Technical Operations Manual.

Field	Value
Product	HORIZON 87 ZENITH Validation Dossier
Generated at	2026-06-08T19:51:03.900Z
Ephemeris source	ZENITH DE441
Reference service	NASA/JPL Horizons API
Frame / observer	J2000 / Earth geocenter / 500@399
Correction convention	NONE - geometric DE441 vector match

SECTION 00

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SECTION 01

Executive Summary

Horizon 87 ZENITH validation compares locally computed CSPICE state vectors against live NASA/JPL Horizons VECTORS rows under matched conventions. This report summarizes the validation dossier generated on 2026-06-08 and records the highest-trust result path for Horizon 87 V1.

IN PLAIN ENGLISH

Horizon 87 asks NASA/JPL Horizons for reference positions, computes the same state vectors locally using its installed DE441 precision package, and reports the difference. In this validation run, all tested vectors passed the selected gates.

Metric	Recorded Value	Interpretation
Fixture count	7	Seven validation epochs spanning J2000, eclipse/stress cases, modern, current, and future dates.
Vector comparisons	63	Nine bodies per fixture: Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune.
Failures	0	No failed comparison rows were recorded.
Maximum Cartesian residual	0.000674349576 m (674.35 um)	Worst total residual across all 63 comparisons.
Maximum angular residual	4.5107e-11 arcsec	Derived from the perpendicular component of the Cartesian residual vector.
Precision status	targeting-1m-passed	Sub-meter certification threshold passed in this dossier.

VALIDATION CLAIM

In this matched validation run, Horizon 87 ZENITH passed 63 / 63 vector comparisons against live NASA/JPL Horizons references. The worst recorded Cartesian residual was 674.35 um. This applies to the recorded fixture set and conventions only.

SECTION 02

Validation Setup & Conventions

The validation is meaningful only because the local runtime and external reference rows use matched target, observer, frame, time scale, and correction conventions. The exported dossier records these conditions so the result can be audited later.

Field	Dossier Value
Source kernel	JPL DE441
Reference service	NASA/JPL Horizons API
Observer	Earth geocenter / 500@399
Output frame	J2000
SPICE frame	J2000
Time input	TDB
Time conversion	Shared Horizons query epoch / ZENITH ET-TDB
Aberration	ZENITH NONE
Reference aberration	Horizons NONE

Field	Dossier Value
Correction convention	NONE - geometric DE441 vector match
Units	km, seconds, arcseconds

Methodology

Each row fetches a live NASA/JPL Horizons VECTORS reference for the same target, observer, frame, TDB time, and selected correction convention, then compares it against the current ZENITH CSPICE state vector at the exact query ET. Angular miss is derived from the perpendicular component of that same Cartesian residual, so the meter and angular fields describe one comparison path.

Reference Epoch Policy

Live Horizons rows are compared at the exact TDB/TT query epoch used to request the vector. Returned JD TDB is recorded for audit only because Horizons may round it for display.

SECTION 03

Precision Asset Package

The validation run was executed with the ZENITH DE441 precision package ready and installed in local browser storage. The package includes split DE441 kernels, a NAIF leap-second kernel, CSPICE WebAssembly runtime files, and metadata.

Package Field	Value
Supported	True
Ready	True
Package version	de441-official-naif-cspice-wasm-v1
Metadata created at	2026-06-04T15:51:05.7088858Z
Installed at	2026-06-08T19:49:41.296Z
Source	https://naif.jpl.nasa.gov/pub/naif/generic_kernels/
Storage estimate	usage 3,342,096,693 bytes / quota 14,079,514,933 bytes

File	Bytes	SHA-256	Source
de441_part-1.bsp	1,651,119,104	13757827f5db41b835a24bbd...7d5b618e	NASA/JPL NAIF
de441_part-2.bsp	1,656,830,976	3abb17dae2d78dd348803775...4d4887f6	NASA/JPL NAIF
cspice.js	66,617	42028bf07b50a7523b48e5b3...a1f9bf72	local CSPICE WASM build
cspice.wasm	1,370,142	73e1c988090b45d66c5dd51b...09e0a28d	local CSPICE WASM build
naif0012.tls	5,407	9c0607d0c8710b3b824867dd...7bcabb5d	NASA/JPL NAIF

OFFLINE PRECISION NOTE

Once the precision assets are installed, ZENITH can compute local DE441 state vectors without re-downloading the kernel package. The live Horizons request remains necessary for external reference validation.

SECTION 04

Fixture Matrix

The dossier contains seven fixture epochs. Each fixture compares nine bodies, producing 63 total vector comparisons. This set checks a mix of baseline, stress, eclipse-related, modern, current, and future dates.

Fixture	UTC Epoch	Vectors	Worst Total Residual
J2000 UTC	2000-Jan-01 12:00:00 UTC	9	Neptune / 674.35 um
Mercury stress UTC	2016-May-09 15:00:00 UTC	9	Uranus / 476.84 um
Penumbral lunar 2020 UTC	2020-Jan-10 19:10:00 UTC	9	Saturn / 238.42 um
Annular solar 2020 UTC	2020-Jun-21 06:40:00 UTC	9	Uranus / 674.35 um
Modern UTC	2025-Jan-01 00:00:00 UTC	9	Saturn / 240.27 um
Current validation UTC	2026-Jun-02 00:00:00 UTC	9	Uranus / 641.96 um
Future UTC	2040-Jun-01 00:00:00 UTC	9	Neptune / 674.35 um

Body Coverage

Each fixture includes Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, and Neptune. The reference observer is Earth geocenter / 500@399, and all comparisons use J2000 with geometric NONE correction.

SECTION 05

Residual Metrics

Horizon 87 reports residuals as components of a single Cartesian comparison path. Angular miss is derived from the perpendicular component of the same residual vector rather than from a separate sky-position path.

Metric	Definition	Meaning
Cartesian residual	$\Delta r = r_{\text{ZENITH}} - r_{\text{Horizons}}$	Vector difference between local and reference state-vector positions.
Total residual	$\text{total_m} = \Delta r * 1000$	3D position difference in meters
Radial residual	$\Delta r_{\text{radial}} = \text{dot}(\Delta r, r_{\text{ref_hat}})$	Residual component along the reference range direction.
Angular-miss residual	$\Delta r_{\text{tangential}} = \sqrt{ \Delta r ^2 - \Delta r_{\text{radial}}^2}$	Perpendicular miss distance, still expressed in meters
Angular residual	$\text{angular_arcsec} = \Delta r_{\text{tangential}} / r_{\text{ref}} * 206264.806247$	Angular equivalent of the perpendicular miss.

Thresholds

Gate	Value	Dossier Result
Certification gate	1 m	Passed: all 63 vectors below 1 m.
Stretch gate	10 m	Passed: all 63 vectors below 10 m.
Goal gate	50 m	Passed: all 63 vectors below 50 m.
Angular limit	0.05 arcsec	Passed within reported maximum angular residual.
Distance limit	1e-07 AU	Passed within reported maximum distance AU.

SECTION 06

Result Summary

All 63 vector rows passed the sub-meter, 10 m, and 50 m gates recorded in the dossier. The worst total Cartesian residual remained under one millimetre.

Summary Field	Value
Vector count	63
Maximum Cartesian residual	0.000674349576174 m (674.35 um)
Maximum radial residual	0.000496197477244 m (496.20 um)
Maximum angular-miss residual	0.000668137432768 m (668.14 um)
Sub-meter pass count	63 / 63
10 m stretch pass count	63 / 63
50 m goal pass count	63 / 63
Certifiable	True

Category	Fixture / Body	UTC	Recorded Value
Worst Cartesian	NEPTUNE / J2000 UTC	2000-Jan-01 12:00:00 UTC	674.35 um total
Worst radial	URANUS / Future UTC	2040-Jun-01 00:00:00 UTC	496.20 um radial
Worst angular miss	URANUS / Annular solar 2020 UTC	2020-Jun-21 06:40:00 UTC	668.14 um angular-miss

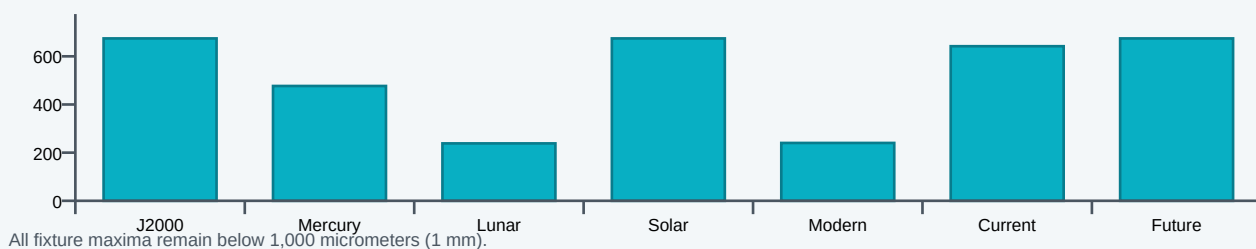
SECTION 06

Result Summary - Per-Fixture Maxima

Fixture	UTC	Worst Total	Worst Radial	Worst Angular Miss
J2000 UTC	2000-Jan-01 12:00:00 UTC	Neptune / 674.35 um	Uranus / 308.13 um	Neptune / 666.01 um
Mercury stress UTC	2016-May-09 15:00:00 UTC	Uranus / 476.84 um	Uranus / 442.46 um	Neptune / 228.48 um

Fixture	UTC	Worst Total	Worst Radial	Worst Angular Miss
Penumbral lunar 2020 UTC	2020-Jan-10 19:10:00 UTC	Saturn / 238.42 um	Saturn / 202.48 um	Uranus / 207.21 um
Annular solar 2020 UTC	2020-Jun-21 06:40:00 UTC	Uranus / 674.35 um	Saturn / 209.01 um	Uranus / 668.14 um
Modern UTC	2025-Jan-01 00:00:00 UTC	Saturn / 240.27 um	Saturn / 225.08 um	Uranus / 191.12 um
Current validation UTC	2026-Jun-02 00:00:00 UTC	Uranus / 641.96 um	Saturn / 478.38 um	Uranus / 504.08 um
Future UTC	2040-Jun-01 00:00:00 UTC	Neptune / 674.35 um	Uranus / 496.20 um	Neptune / 658.90 um

Maximum total Cartesian residual by fixture (micrometers)



SECTION 07

Timing Offset Diagnostic

The dossier includes an ET microsecond-offset sweep around the worst Cartesian case. This diagnostic checks whether shifting the comparison epoch by small amounts would reduce the residual, which could indicate a hidden timing offset.

Field	Value
Status	complete
Target	NEPTUNE / J2000 UTC
UTC	2000-Jan-01 12:00:00 UTC
Sweep range	-100 to 100 microseconds
Step	5 microseconds
Baseline residual	674.35 um
Best offset	0 microseconds
Best residual	674.35 um
Interpretation	Best residual remains near the unshifted epoch; no obvious microsecond timing offset is indicated.

Offset (us)	Total Residual	Radial	Angular Miss
-100	3.53 m	1.161 m	3.34 m
-50	1.77 m	58.05 cm	1.67 m
-25	88.37 cm	29.01 cm	83.48 cm
-10	35.40 cm	11.57 cm	33.46 cm
-5	17.76 cm	5.77 cm	16.80 cm
0	674.35 um	105.72 um	666.01 um
5	17.53 cm	5.82 cm	16.54 cm
10	35.17 cm	11.62 cm	33.20 cm
25	88.14 cm	29.06 cm	83.22 cm
50	1.76 m	58.10 cm	1.67 m
100	3.53 m	1.161 m	3.33 m

DIAGNOSTIC CONCLUSION

The best residual remains at the unshifted epoch. The sweep therefore does not indicate an obvious microsecond timing offset in the validation pipeline.

SECTION 08

Legacy Metadata Note

The exported JSON contains a `strictMode` field. In this dossier it is `false`. This field should not be interpreted as the validation mode or as a weakening of the recorded validation result.

STRICT MODE FIELD

In Horizon 87 V1, `strictMode` refers to an older runtime failsafe that could prevent ZENITH from running if validation did not pass. It is not the validation criterion used in this report. The validation result is determined by the recorded Horizons comparison rows, pass/fail counts, selected residual thresholds, and precision summary.

Validation Criterion	Dossier Evidence
Overall status	ok=True, status=passed
Reference source	live NASA/JPL Horizons API VECTORS response
Local runtime	ZENITH DE441 active
Failed count	0
Selected gate	50 m
Precision pass	subMeter=True, stretch10m=True, goal50m=True

SECTION 09

Claim Boundaries

This report records a high-trust validation result, but the claim remains bounded by the validation setup. The report does not convert every Horizon 87 feature into a universal state-vector certificate.

Area	Allowed Claim	Boundary
ZENITH validation	Matched local DE441/CSPICE vectors agreed with live Horizons references within the recorded residuals.	Applies to the specific fixture set, bodies, frames, observer, time scale, and NONE correction convention.
Visualizer	Uses the active ephemeris path to render solar-system geometry.	Rendered scenes are not independent validation certificates.
Observer tools	Provide geometric sky guidance using altitude, twilight, elongation, and location filters.	No weather, terrain, building, extinction, optics, or eyesight guarantee.
Mission Lab	Provides preliminary Lambert transfer-window screening.	Not a spacecraft navigation product; no finite-burn, launch vehicle, gravity-assist, or operations model.
Micrometer-scale statement	Valid as a run-specific observation for this dossier.	Not a universal promise for every object, date, display mode, scanner, or future runtime state.

PUBLIC WORDING

Recommended wording: 'In a 63-vector matched validation run against live NASA/JPL Horizons VECTORS references, Horizon 87 ZENITH passed all tested comparisons with a maximum Cartesian residual of 674.35 um under recorded conventions.'

SECTION 10

Appendix & References

The source artifact for this report is the exported ZENITH validation dossier listed below. External reference ecosystems are named for reproducibility and context; Horizon 87 is not affiliated with, endorsed by, or certified by NASA, JPL, NAIF, or any space agency.

Item	Value
Source JSON	horizon87-validation-2026-06-08T19-51-19-935Z.json
Product	HORIZON 87 ZENITH Validation Dossier
Generated at	2026-06-08T19:51:03.900Z
Package version	de441-official-naif-cspice-wasm-v1
Installed files	de441_part-1.bsp, de441_part-2.bsp, naif0012.tls, cspice.wasm, cspice.js, zenith_metadata.json
Missing files	None

Resource	URL
NASA/JPL Horizons API Documentation	https://ssd-api.jpl.nasa.gov/doc/horizons.html
NASA/JPL Horizons System	https://ssd.jpl.nasa.gov/horizons/
NAIF CSPICE SPK Required Reading	https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/spk.html
NAIF SPICE Kernel Required Reading	https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/kernel.html